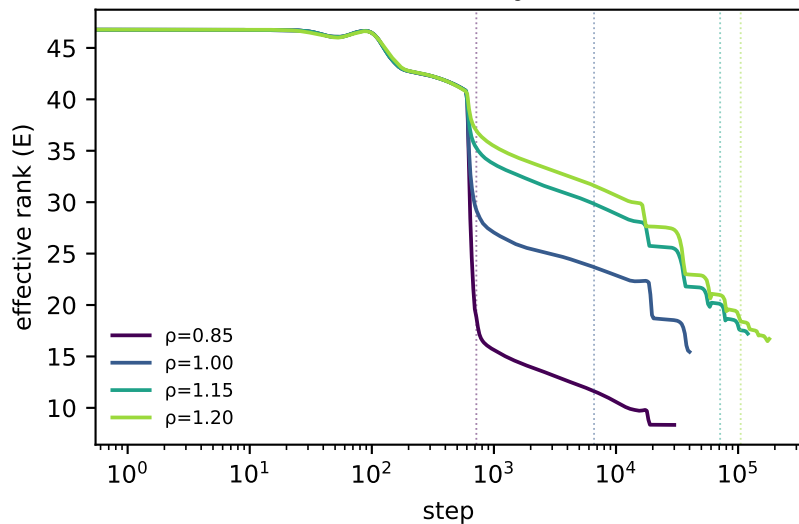
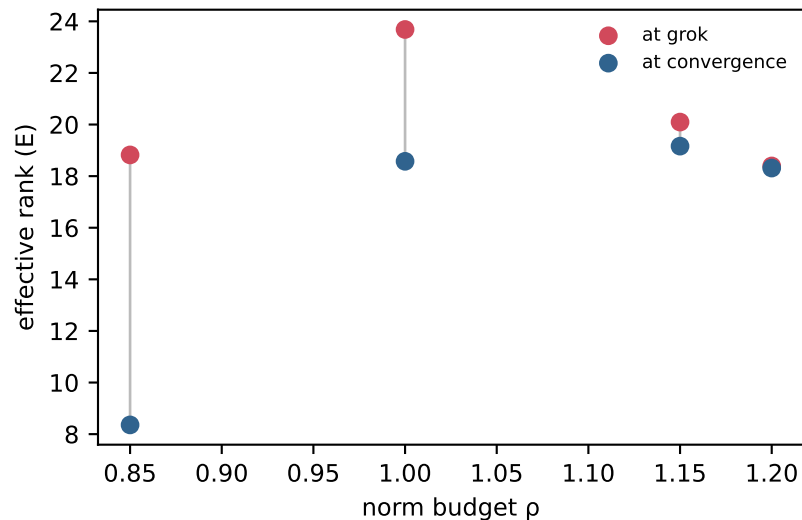


Compression-clock audit — transformer no-LN (real data, Paper B run)

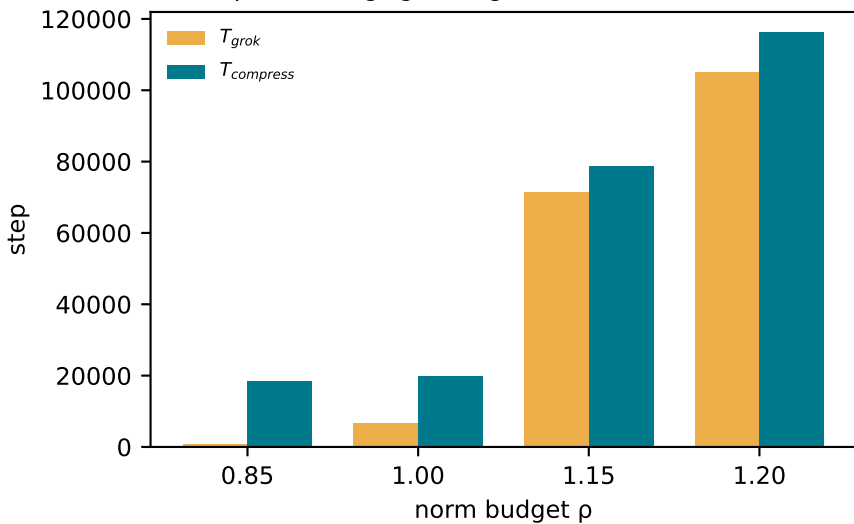
A eff_rank trajectories (dotted = T_{grok})



B at-grok is not converged



C compression lags grokking



D boundary gate (red = excluded)

